IMPERIAL

Magnitude-Preserving Autoencoders

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High-dimensional data with low-dimensional structure

High-dimensional data in fields such as climate, finance, insurance and astronomy possess low-dimensional structure. A single observation may live in hundreds or thousands of dimensions, yet concentrate near a low-dimensional manifold. Issues such as computational efficiency and ambient measure collapse motivate dimension reduction and representation learning techniques.

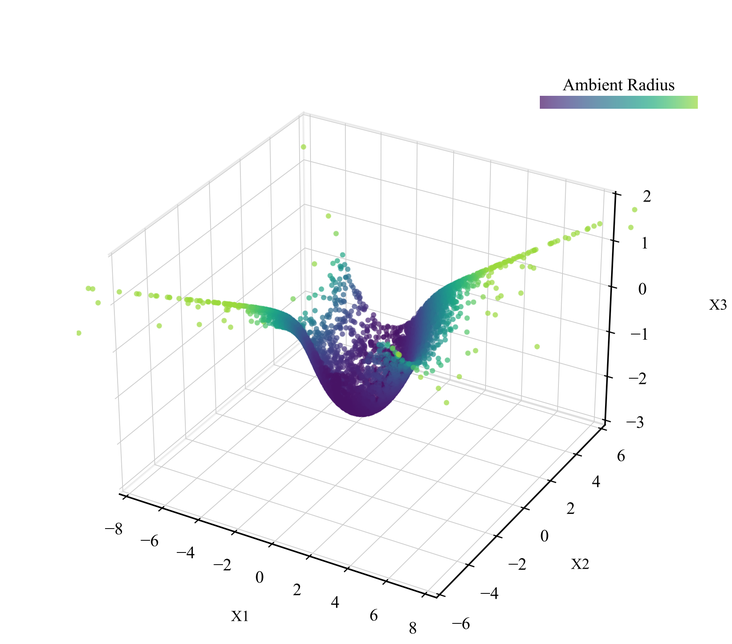


Figure 1: Observations on a curved 2-surface in $\mathbb{R}^3$.

The events of greatest practical interest, such as climate disasters, market crashes and insurance losses, often involve the most **extreme** observations in the tails of the distribution. A useful latent representation should transport the data to a lower dimensional space while retaining a notion of ambient space extremeness.

Standard autoencoders do not preserve magnitude

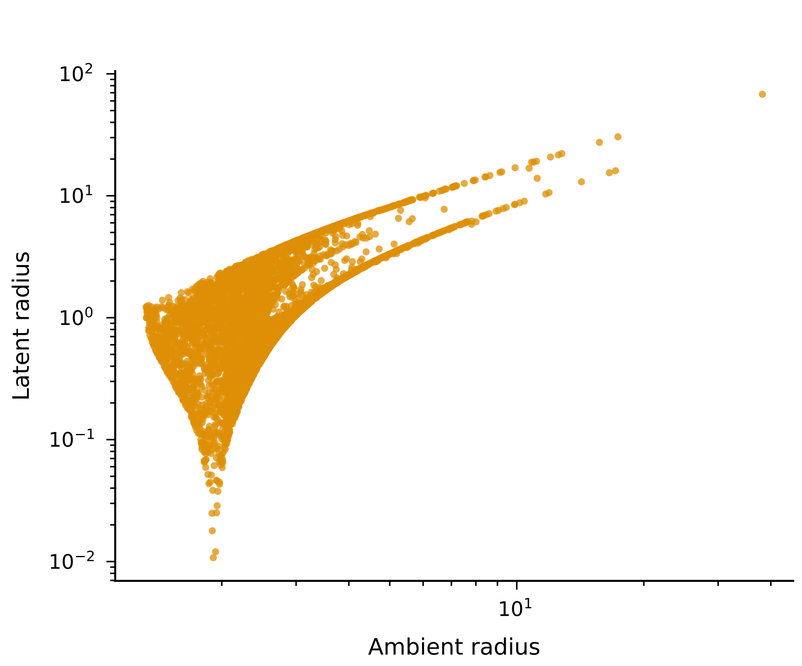


Figure 2: Latent $\|Z\|$ vs ambient $\|X\|$ radius for data encoded with a standard AE: $\|X\|$ does not have a clear correspondence to $\|Z\|$.

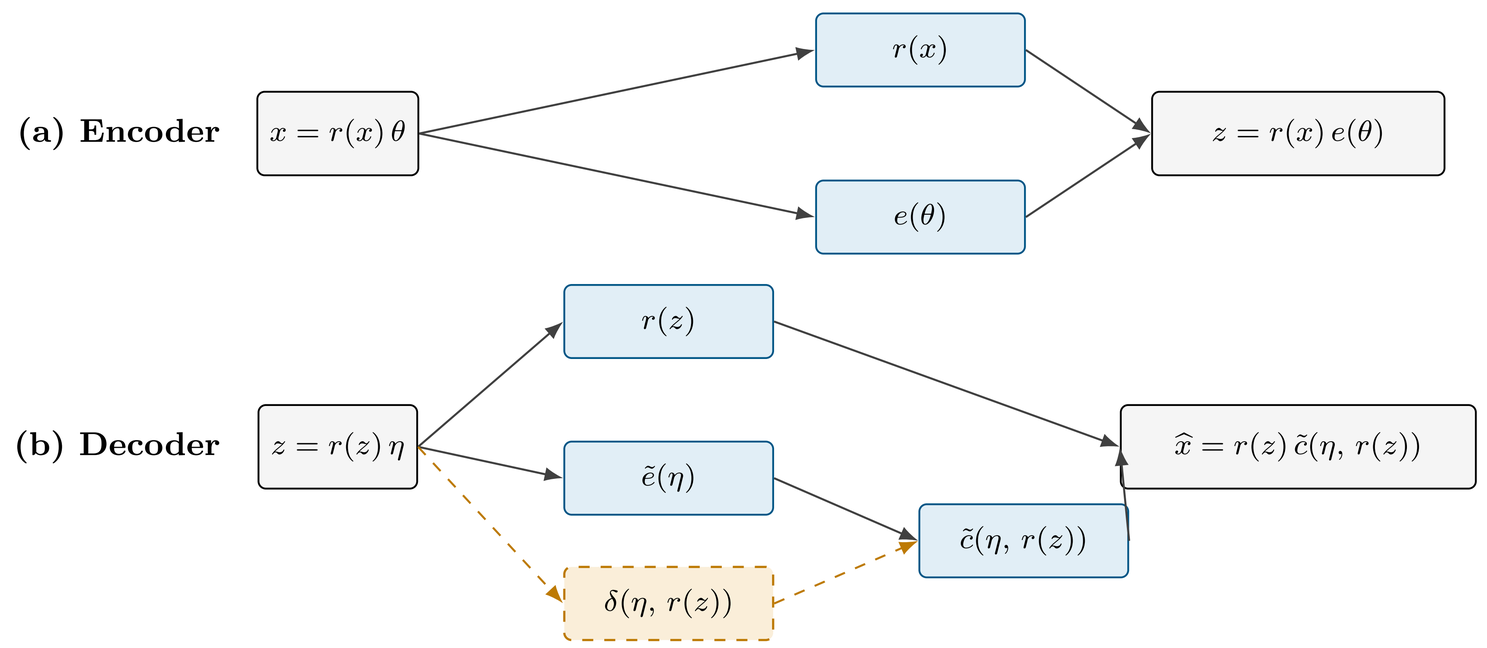
An unconstrained autoencoder distorts radial scaling: $\|X\|$ is not reliably mapped to $\|Z\|$. This means that analysis of magnitude in latent space, such as latent tail index estimates, loses meaning with respect to the ambient data.

Method: Homogeneous Autoencoder (HAE)

Decompose each ambient point as $x = r(x)\,\theta$ and transform only the angular component leaving the radius preserved. We construct an **encoder** which is 1-homogeneous and a **decoder** which is asymptotically 1-homogeneous:

$$f(x) = r(x)\,e(\theta), \qquad h(z) = r(z)\,\tilde{c}(\eta, r(z)),$$

where $\tilde{c}$ renormalises $\tilde{e}(\eta)+\delta(\eta, r(z))$ onto the unit sphere and $\sup_\eta \|\delta(\eta, r(z))\| \to 0$ as $r(z) \to \infty$. The construction guarantees $r(f(x)) = r(x)$ and $r(h(f(x))) = r(x)$ exactly.



HAE with ERA5 atmospheric reanalysis data

ERA5 surface (10m) wind u_{10} over southwest England and Wales. $D{=}195$ grid cells, $m{=}8$. HAE versus a parameter-matched standard AE and PCA for comparison.

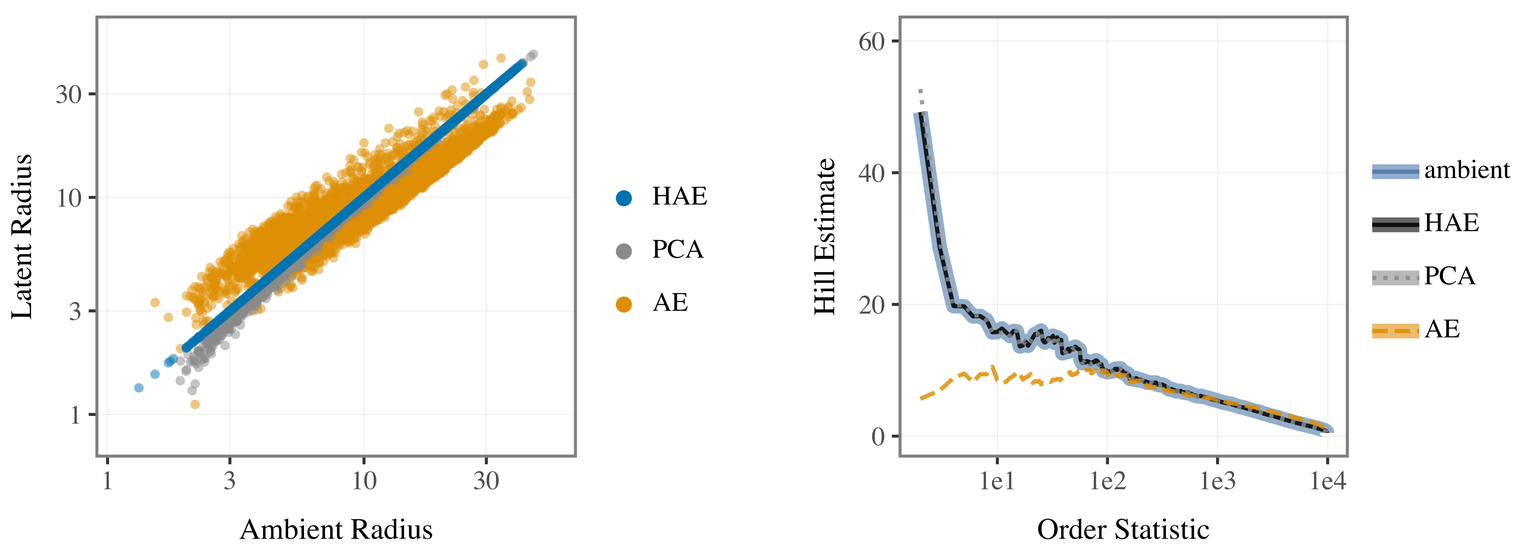


Figure 3: Left: latent vs ambient radius on u_{10} (AE drifts into a cloud). Right: latent Hill estimator (HAE matches ambient, AE drifts).

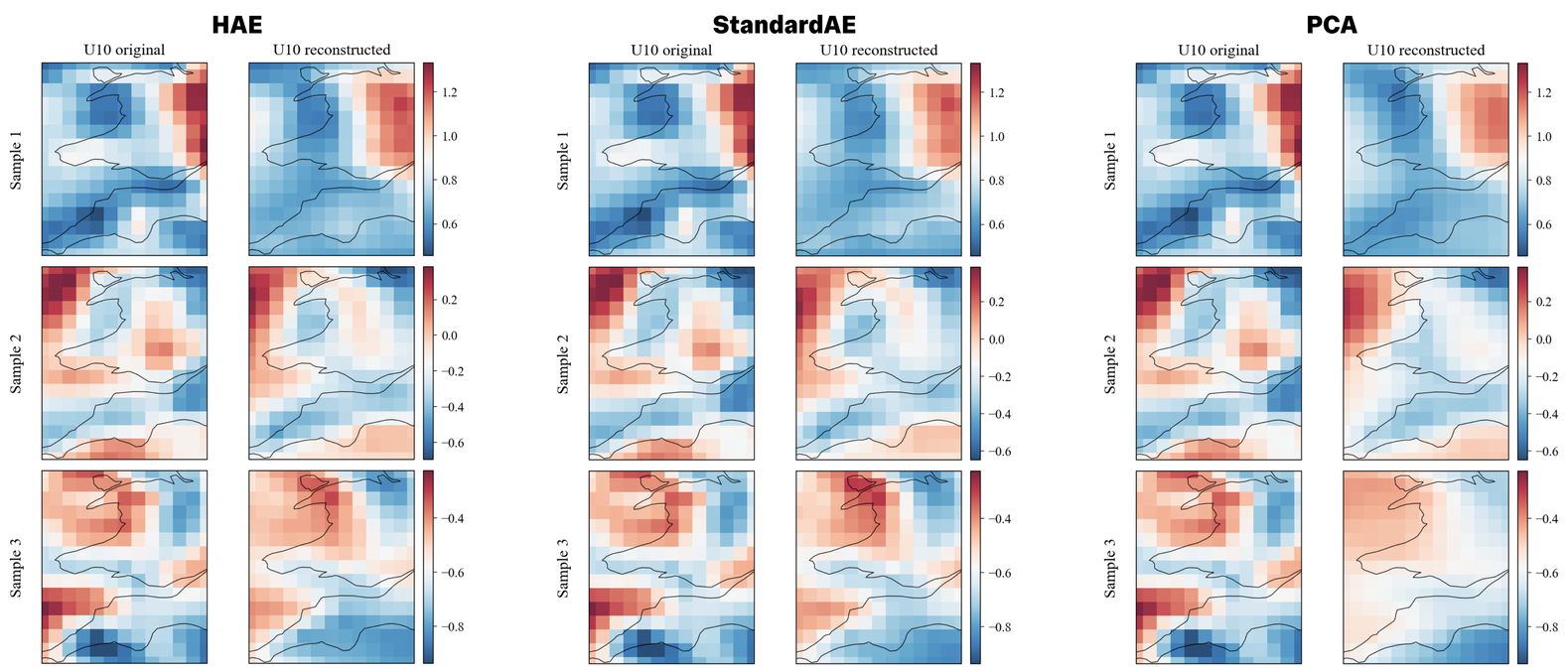


Figure 4: u_{10} wind-field reconstructions: original (left) vs reconstructed (right) per model.

Reconstruction MSE: HAE 0.0096, AE 0.0089, PCA 0.014.
Latent Hill drift of 0 for HAE compared to 0.15 for the AE.